A Mini Literature Review on Low-Cost LoRa-Based Weather Stations for Remote Environmental Monitoring

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***Abstract*—Collecting weather data in large agricultural fields, mountains, and remote areas is difficult due to the lack of Wi-Fi and unstable or expensive cellular networks (2G/3G/4G). Long Range (LoRa) and LoRaWAN technologies address this issue by enabling multi-kilometer wireless transmission with ultra-low power consumption and no per-node telecommunication subscription fees. This mini literature review analyzes six selected studies on LoRa-based environmental monitoring systems across four main themes: (1) wireless communication and range in complex terrains, (2) microcontrollers, transceivers, and sensor selection, (3) power management and offline data reliability, and (4) network topologies and data visualization platforms. We then compare these studies with our proposed "Weather Station x LoRa" station node, which integrates an ESP32 microcontroller, an MKE-M24 Ra-01SH LoRa module (915 MHz), a DHT11 sensor, a BH1750 light sensor, a Grove soil moisture sensor, a DS1307 real-time clock (RTC), and a MicroSD card reader, while noting current sensor accuracy and budget limitations.**

***Index Terms*—Internet of Things (IoT), LoRa, LoRaWAN, Weather Station, Remote Environmental Monitoring, Smart Agriculture.**

I. Introduction

Continuous weather monitoring is important for weather forecasting, agriculture, public health, and tourism [1], as well as climate research and natural disaster early warning [2], [3]. In agriculture, farmers need continuous information about air temperature, relative humidity, soil conditions, and solar radiation to monitor fields and manage crops effectively [4], [5]. However, deploying automatic weather stations in remote rural areas or mountainous terrains faces major financial and technical barriers [3], [5], [6]. Conventional commercial weather stations can cost many thousands of dollars, restricting wide-scale deployment in developing regions [3], [6].

Furthermore, standard wireless communication technologies struggle in isolated locations [4], [5]. Short-range modules such as Wi-Fi, Bluetooth, and Zigbee have limited coverage—typically less than 100 meters [1], or below 1,000 meters for general local/personal area networks [5]—while technologies such as Bluetooth, Wi-Fi, Zigbee, and GSM can require significant power consumption [2], [4]. Conversely, cellular networks (GSM/2G/3G/4G) often suffer from weak reception or coverage "blank spots" in mountains and rural farmlands [3]–[5], while requiring recurring telecommunication subscription fees [3], [5], [6].

To answer the core research question—*"How can we measure environmental conditions in remote areas?"*—researchers have adopted Low-Power Wide-Area Network (LPWAN) technologies, specifically LoRa and LoRaWAN [2], [3], [5]. Operating in unlicensed Industrial, Scientific, and Medical (ISM) radio bands (433 MHz, 868 MHz, and 915 MHz), LoRa transmits data over several kilometers without internet access at the field node or per-node subscription charges [5], [6] (although gateways using cellular backhaul may still incur minor annual SIM fees [3]). Its low power consumption also allows stations to run independently on solar panels and rechargeable batteries, as implemented in [3], [5] and recommended for future work in [2]. This paper reviews six selected studies [1]–[6] to establish the technical foundation for our "Weather Station x LoRa" project.

II. Thematic Literature Review

A. Wireless Communication and Range in Complex Terrains

LoRa technology uses Chirp Spread Spectrum (CSS) modulation [5], and studies report that signal demodulation is possible under noise levels below 20 dB [1], [2], with receiver sensitivity reaching as low as -137 dBm [3]. Across the literature, Pietrosemoli et al. [6] used the 868 MHz band in Europe to extend short-range (25–100 m) 433 MHz commercial weather stations into LoRaWAN nodes, whereas Murdyantoro et al. [1], Hidayat et al. [4], and Lopez et al. [5] operated at 915 MHz in Indonesia and Peru.

TABLE I
COMPARISON OF REVIEWED LORA/LORAWAN WEATHER STATION SYSTEMS AND OUR PROPOSED SYSTEM

Study / System	MCU & LoRa Hardware	Frequency & Range	Measured Parameters (Sensors)	Power & Local Storage	Server & Visualization
Dajnowski et al. (2024) [2]	Arduino MCU, Dragino LoRa Shield, LPS8 Gateway	868 / 915 MHz; 1–10 km (theoretical LoS, not field-tested)	Temp, humidity (DHT11/22, BME280), pressure, wind speed, rain, gas (MQ-135)	Battery only (rapid drain reported); No local storage	TTN, ThingSpeak (MATLAB), custom Flask/HTML GUI
Hidayat et al. (2019) [4]	ESP32 (TTGO LoRa32 V2) for node & gateway	915 MHz (P2P); 200 m (obstacle) to 800 m (LoS)	Temp, humidity (DHT22), solar radiation (TSL2561)	Battery with Deep Sleep; RTC DS3231 + SD Card	Custom cloud (agri-eye), phpMyAdmin (.csv/.xlsx)
Zennaro et al. (2025) [3]	Barani MeteoHelix, SenseCAP, Milesight UG67	LoRaWAN; 260 m–1.5 km (buried); 25–40 km cell link (planned)	Temp, humidity, pressure, solar rad (MeteoHelix), wind, rain, soil (S2105), tilt, water level	305 Wp solar + 100 Ah battery; 2–8 GB flash (DTG)	Embedded LNS, MQTT (QoS 2), Telegraf, InfluxDB, Grafana
Lopez et al. (2025) [5]	ESP32 WROOM 32, E220-900T22D-V1.2 LoRa	915 MHz; 5 km (obstacles) to 10 km (LoS)	Air temp/humidity (CWT-TH04S), soil temp (DS18B20), solar rad, wind, pressure	36–37 W solar + LiPo, sleep (95–120 s) + relay; No SD	Custom Laravel server, REST API, MySQL, Grafana
Pietrosemoli et al. (2019) [6]	BSF32 / Lopy4 (ESP32 + SX1276), RXB6 433 MHz	433 MHz (in) to 868 MHz (LoRaWAN out)	Temp, humidity, wind, rain (AcuRite / La Crosse WS), air pressure (BMP180)	BSF32 (2000 mAh Li-ion, 144 μA) / Lopy4 (48 Wh, 32.8 μA); No SD	TTN community network and cloud server
Murdyantoro et al. (2019) [1]	Arduino UNO Rev3, LoRa Shield, LG01-S Gateway	915 MHz; 5–10 km (theoretical, not field-tested)	Temp, humidity (DHT11), pressure (BMP180), rainfall (raindrop), wind speed	Power bank; No local storage	LG01-S (Wi-Fi client) directly to ThingSpeak
Proposed System (Group 2)	**ESP32, MKE-M24 Ra-01SH LoRa, 3.5 dBi Antenna**	**915 MHz; Multi-km target without internet**	**Air temp & humidity (DHT11), light intensity (BH1750 I2C), soil moisture (Grove)**	**Solar in problem statement (not in component list); RTC DS1307 + MicroSD**	**Timestamped local MicroSD logging & LoRa transmission**

While Murdyantoro et al. [1] and Dajnowski et al. [2] only noted theoretical LoRa ranges of 5–10 km and 1–10 km without conducting field distance tests, other studies evaluated actual range in complex terrains. In Yogyakarta, Hidayat et al. [4] found that a 915 MHz peer-to-peer (P2P) link reached 800 m in clear Line-of-Sight (LoS, Zone D) with an average Received Signal Strength Indicator (RSSI) of -114 dB and 20% packet loss. However, a 39 m high building blocking Zone B reduced the range to 200 m with an average RSSI of -124 dB and 80% packet loss [4]. Dajnowski et al. [2] similarly noted that clear LoS installation without building interference is necessary to prevent rapid signal loss.

In high-altitude mountains, Lopez et al. [5] tested a 915 MHz system in the Peruvian Andes (3,800–4,000 m altitude), achieving 100% packet delivery over a 10 km LoS link in open terrain (Muñani) and a 5 km link in agricultural terrain with scattered houses (Juliaca). In Kyrgyzstan's mountains, Zennaro et al. [3] received data from buried soil sensors at distances of 260 m to 1,500 m, noting that even longer ranges could be achieved because LoRa decodes signals down to -137 dBm. In addition, at their Adygene Lake site (3,600 m altitude, where installation was still in progress), desk analysis identified some spots to link the gateway via cellular backhaul to mobile towers 25–40 km away [3].

B. Microcontrollers, Transceivers, and Sensor Selection

The reviewed studies employ two main groups of microcontroller units (MCUs) alongside commercial stations:

- **Arduino Platforms:** Dajnowski et al. [2] used an Arduino MCU with a Dragino LoRa shield and an LPS8 gateway, while Murdyantoro et al. [1] used an Arduino UNO Rev3 with a LoRa shield and an LG01-S gateway. However, Dajnowski et al. [2] reported that connecting multiple sensors stretched the Arduino MCU's memory capacity and hardware limits to the maximum.
- **ESP32 and Commercial Platforms:** Hidayat et al. [4] used ESP32-based TTGO LoRa32 V2 boards for both the sensing node and gateway. Pietrosemoli et al. [6] upgraded their BSF32 transponder to an ESP32-based Lopy4 (with an SX1276 transceiver), noting that the Lopy4 offers larger memory and a faster clock speed, although it is more expensive. Lopez et al. [5] paired an ESP32 WROOM 32 with an E220-900T22D-V1.2 LoRa module, building a complete prototype for about $320 USD. For pilot deployments in Kyrgyzstan, Zennaro et al. [3] used IP67 Milesight UG67 gateways and Barani MeteoHelix IoT PRO stations designed to meet WMO quality standards (reported as starting at or costing below $1,000 USD, whereas traditional stations cost many thousands of dollars, with one reported conventional station in At-Bashy costing $80,000 USD).

All six studies measure air temperature and humidity [1]–[6], often using low-cost DHT11 [1], [2] or DHT22 [2], [4] sensors. Air pressure is measured via BMP180 [1], [6] or BME280 [2], while light and solar radiation are measured by TSL2561 [4], SN-300AL-RA-N01 [5], or Barani MeteoHelix

IoT PRO [3] (after testing a SenseCAP S1000 [3]). For soil monitoring, soil temperature is measured with a DS18B20 [5], and soil moisture and temperature with a SenseCAP S2105 [3]. Several systems also include anemometers and wind vanes [1], [2], [5], [6], rain sensors or gauges [1]–[3], [6], MQ-135 gas sensors [2], and ultrasonic water level or tilt sensors [3].

C. Power Management and Offline Data Reliability

Off-grid operation requires strict energy management. Dajnowski et al. [2] observed rapid battery depletion when running multiple sensors continuously without solar panels. In sub-zero Andean temperatures, Lopez et al. [5] powered the node with a 36–37 W solar panel and reduced battery drain with a 95–120 s MCU sleep cycle and a relay to cut sensor power between readings. Similarly, Hidayat et al. [4] used ESP32 Deep Sleep, and in Pietrosemoli et al. [6], the Lopy4 transponder reduced sleep current to $32.8\,\mu$A (compared to $144\,\mu$A on the BSF32, which lasted 123 days), estimating a 414-day life with a 48 Wh battery pack at 15-minute intervals.

To prevent data loss when rural internet backhaul fails, only two studies implemented local hardware storage: Hidayat et al. [4] added a Real-Time Clock (RTC DS3231) and an SD card on the gateway to log timestamped packets during Wi-Fi outages, while Zennaro et al. [3] used a Disruption Tolerant Gateway (DTG) with 2–8 GB internal flash to store over 2 years of messages during cellular outages.

D. Network Topologies and Data Visualization Platforms

Regarding server integration, Dajnowski et al. [2] used The Things Network (TTN) with ThingSpeak (noting that ThingSpeak limits each channel to 8 data fields), Pietrosemoli et al. [6] used TTN, and Murdyantoro et al. [1] configured an LG01-S gateway as a Wi-Fi client to send data directly to ThingSpeak via HTTP API keys. Conversely, Hidayat et al. [4] used a P2P link connected to a custom cloud server (*agrieye*) with phpMyAdmin, Lopez et al. [5] (whose system is termed LoRaWAN but uses a direct E220 transmitter-to-receiver link) connected to a Laravel server, MySQL, and Grafana, and Zennaro et al. [3] used an embedded LNS, MQTT, Telegraf, InfluxDB, and Grafana.

III. Synthesis and Alignment with Proposed System

Table I compares the six reviewed systems with our proposed "Weather Station x LoRa" station design:

- **Processing & Long-Range Link:** Following the use of ESP32 boards in [4]–[6], our station uses an **ESP32 MCU** to avoid the memory limits of the Arduino MCU reported in [2]. Paired with an **MKE-M24 Ra-01SH LoRa module** and a **915 MHz 3.5 dBi antenna**, the node is designed to transmit meteorological data over kilometers without internet or periodic telecom fees, operating in the same 915 MHz band used in [1], [4], [5]. While our problem statement mentions independent solar operation, our current component list does not include solar charging hardware.
- **Agricultural Sensing:** Our station measures four parameters: air temperature and humidity via a **DHT11 sensor** (the same sensor model used in [1], [2]), light intensity via a **BH1750 I2C sensor** (following the inclusion of light/solar sensors such as TSL2561 in [4]), and soil moisture using a **Grove Soil Moisture sensor** (aligning with the soil moisture monitoring in [3] and soil monitoring in [5]).
- **Offline Data Logging:** Unlike designs without local backup [1], [2], [5], [6], our hardware includes an **MKE-M09 RTC DS1307** module and a **MicroSD card reader** to support timestamped local data logging, adopting the offline storage principle demonstrated in [3], [4].
- **Limitations:** While [6] notes that consumer-grade stations cannot match the performance of professional stations, [1] attributes reading differences to sensor types and accuracy levels. Due to budget constraints, our prototype has high sensor measurement error and currently lacks wind speed, wind direction, and ultraviolet (UV) sensors, which will be addressed in future upgrades.

IV. Conclusion

The reviewed literature supports the feasibility of sub-GHz LoRa technology (such as 915 MHz) for monitoring environmental conditions in remote and mountainous areas without Wi-Fi or cellular coverage, provided that line-of-sight placement and power management are carefully considered. By combining an ESP32 MCU, a 915 MHz Ra-01SH LoRa transceiver, environmental sensors (DHT11, BH1750, Grove soil moisture), and local logging hardware (RTC DS1307 and MicroSD reader), our "Weather Station x LoRa" prototype builds upon proven practices from the reviewed studies to establish a practical foundation for remote environmental monitoring.

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